Sentiment Analysis of Farsi (Dari) Social Media Text

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1 Introduction

Sentiment analysis is the computational study of people's opinions, emotions, and attitudes expressed in text. Approximately 85% of digital data today is unstructured and noisy, consisting of social media posts, chat logs, customer reviews, and emails [1]. These texts often include dialectal variations, misspellings, emojis, and informal phrasing. For instance, the Dari phrase سلام خوبی؟ might appear as its Latin transliteration "Salam Khobi".

With the digitalization of personal and organizational communication, sentiment analysis has become increasingly crucial across domains. While English has benefited from extensive research and robust models, low-resource languages like Persian (Dari) have lagged behind. Transformer-based models such as ParsBERT [2] have shown promise in addressing Persian NLP tasks, yet challenges remain due to limited labeled data and benchmark resources.

This study addresses these gaps by collecting and annotating a diverse dataset of Dari comments from YouTube and Telegram, capturing dialectal and contextual variety across Afghan speakers. A subset of the data was manually labeled into eight emotion categories—based on guidelines from the PersianEmo [3], with the addition of a "hope" class. The fine-tuned ParsBERT model was then evaluated for performance, revealing insights into the limitations of zero-shot methods and the necessity of task-specific fine-tuning.

2 Literature Review

Recent sentiment analysis research increasingly leverages transformer architectures and attention mechanisms. Tolebay et al. [4] propose a hybrid model combining RoBERTa and DistilBERT with classical classifiers, achieving improved accuracy in social media sentiment classification for low-resource languages. Yal. [5] introduced an LSTM with multi-head attention, outperforming standard LSTM baselines by up to 12% in emotion classification.

This reflects a shift away from lexicon-based and traditional machine learning approaches toward deep learning frameworks optimized for contextual understanding. Within the Persian NLP domain, ParsBERT [2]—a RoBERTa-based transformer fine-tuned on Persian corpora—has been validated as state-of-the-art for sentiment analysis, POS tagging, and NER. Several studies [6, 7] further confirm its robustness when fine-tuned on domain-specific datasets.

Nonetheless, researchers have identified persistent challenges in Persian and Dari sentiment analysis, such as poor generalizability of zero-shot models, sarcasm, and the lack of labeled emotion data [8, 9]. Social media adds additional complexity through multilingual content, spelling variation, and informal tone.

To address these limitations, the SuperEmotion dataset [10] and PersianEmo [3] offer valuable framework for annotating fine-grained emotions, which my study extends by incorporating a "hope" category. This supports broader emotion modeling in socio-political and cultural contexts relevant to Dari speakers.

My approach contributes by combining domain-aware data collection with culturally-informed annotation, advancing both dataset quality and model relevance for Persian (Dari) sentiment analysis.

3 Methodology

3.1 Data Collection

Telegram Data Collection:

I manually selected Telegram channels by directly searching for highly active groups in my Telegram app. One specific group I targeted had approximately 8,000 members and was extremely active, generating over 5,000 new messages daily. This particular group served as a linguistic and cultural bridge, hosting mainly Afghan members residing in Iran, resulting in a diverse mix of Persian and

Dari dialects and expressions. Additionally, I used keyword-based searches focused on topics related to Afghanistan and its major cities to identify other relevant channels. This method enabled me to gather over 23,714 messages, which I stored in JSON format to maintain structural integrity for later processing.

YouTube Data Collection:

To further enhance my dataset, I manually selected YouTube channels known for regularly posting Farsi (Dari) content. These channels included diverse regions of Afghanistan, as well as diaspora communities in places like Quetta, Pakistan. My selection encompassed political channels such as Kaktos, cultural platforms like Nama Rasana, and regional channels like Bamyan Man, Voice of Behsood, People Media, and Reza & Fatima. The global dispersion of commenters provided a broad spectrum of dialects, perspectives, and cultural insights.

3.2 Data Preprocessing

The dataset underwent comprehensive preprocessing to ensure quality and relevance. First, I manually removed duplicate and empty comments. Then, I applied multiple text-cleaning steps: removing URLs, hashtags, user mentions, and non-Persian characters. Additionally, comments with fewer than four words were excluded, as they lacked sufficient context. I normalized and tokenized the text using Hazm, a Persian NLP library, and removed common Persian stopwords to enhance clarity. Each cleaned comment included normalized text and token counts, and this processed data was saved to a CSV file for analysis.

3.3 Dataset Construction

After cleaning, I manually merged the Telegram and YouTube datasets into a unified CSV dataset, carefully standardizing fields and retaining essential information such as channel names, text content, and publication dates. This resulted in a comprehensive dataset suitable for sentiment analysis. To illustrate the scale and composition of the final dataset, I analyzed several core files individually:

- **BV.csv:** 17,963 comments across 637 unique videos with 3,760 distinct authors. A few messages and author fields were missing.
- **Kaktos.csv:** A large corpus of 198,511 messages with 58,959 unique authors. 29 texts were missing, but coverage was wide.
- **NR.csv:** 87,788 entries with 74,308 unique texts and 32,543 unique authors, collected from a major Afghan community channel.
- **PeopleMedia.csv:** A smaller but focused dataset of 1,084 comments across 84 videos.
- **RezaFatima.csv:** A diverse set of 181,514 records from a channel combining cultural and regional themes.
- **telgram_dateset.csv:** Contains 24,192 Telegram messages across manually selected and keyword-searched groups. All records are complete and timestamped.
- **Cleaned_Dataset.csv:** The final cleaned dataset includes 373,105 entries, where each message is normalized, tokenized, and stopwords are removed. The average token count per message is approximately 15.6.

These statistics demonstrate not only the scale of my work but also the balance between diversity and data quality.

3.4 Labeling Method and Emotion Classes

As part of the emotion labeling process, I explored several approaches before deciding on a semi-supervised strategy. The dataset consisted of cleaned Persian (Dari) user comments, along with metadata such as timestamps and token counts.

Initially, I experimented with zero-shot classification using the `joeddav/xlm-roberta-large-xnli` model. However, its performance on Dari text proved to be inconsistent and often inaccurate. Based on this, I concluded that zero-shot methods are not reliable for emotion classification in this context. Instead, I chose to use the `HooshvareLab/bert-base-parsbert-uncased` model (ParsBERT). After successfully authenticating with Hugging Face via a personal API token, I loaded the tokenizer and model in Google Colab. A test input such as سلام دنیا؟ confirmed the model was functioning properly.

I adopted a semi-supervised labeling method: I labeled manually a small subset of my data, which was then used to fine-tune ParsBERT. And later on i used fine-tuned model to label the remaining dataset automatically.

Eight emotion categories were defined to guide both manual and automatic labeling:

- happiness
- sadness
- anger
- fear
- disgust
- surprise
- neutral
- hope

After looking and reading the comments and messages I added intentionally **"hope"** — it reflects forward-looking, optimistic comments that don't quite fall under happiness but indicate encouragement or aspiration. Although the text is in Persian, I used English labels to maintain compatibility with machine learning libraries and Hugging Face tools. If needed, the labels can be translated back into Persian for UI display or reporting.

3.5 Data Preparation for Manual Labeling

To prepare the dataset for manual emotion labeling, I performed proportional sampling from six key files: `BV.csv`, `kaktos.csv`, `NR.csv`, `PeopleMedia.csv`, `RezaFatima.csv`, and `telgram_dateset.csv`. Approximately 4,000 rows were sampled, with the number from each source based on its relative size to ensure balanced representation.

The sampled comments underwent thorough preprocessing, including the removal of non-Persian characters (such as emojis and Latin script), normalization of spacing, and exclusion of short comments with fewer than three tokens. I also eliminated duplicate entries and dropped unnecessary columns like `channel`, `chat_id`, `msg_id`, and `date`. A new column called `label` was added to support manual annotation. The resulting dataset was saved and uploaded to Google Drive for annotation via Google Sheets.

Here is a breakdown of how many comments were sampled from each source:

- BV: 176 rows
- Kaktos: 1,942 rows
- NR: 859 rows

- PeopleMedia: 11 rows
- RezaFatima: 1,776 rows
- Telegram: 237 rows

After manually labeling approximately 4,000 samples across eight emotion categories, I reviewed the labels to ensure consistency. Some emotions like *fear* and *disgust* were less frequent, which reflects their natural rarity in everyday online comments. Still, the balance was sufficient for training.

The final annotated dataset was exported from Excel and further cleaned. I standardized column names, removed rows with missing values, mapped emotion labels to numeric IDs, and retained only essential columns: cleaned text, label, label ID, and source. This refined dataset saved and is now ready for fine-tuning ParsBERT on a larger scale.

4 Results

To train a Persian emotion classification model, I fine-tuned the `HooshvareLab/bert-base-parsbert-uncased` model using a manually labeled dataset covering eight emotion categories.

After data preprocessing and applying a stratified 80/20 split for training and validation, I used the Hugging Face `Trainer` API in PyTorch format with tokenized inputs. The training was performed on Google Colab using the following hyperparameters:

- Epochs: 3
- Batch size: 8
- Learning rate: 2e-5
- Evaluation metrics: Accuracy and weighted F1-score

The model exhibited steady improvement in training loss across epochs, while validation loss remained relatively stable. The best performance was achieved at epoch 3:

- Training Loss: 0.8019
- Validation Loss: 1.4159
- Accuracy: 53.89%
- F1 Score: 52.17%

Although the dataset included label imbalance and linguistic complexity, these results demonstrate a meaningful learning trend. The model is now prepared for further evaluation, per-class analysis, and potential use in semi-supervised labeling of the remaining 400,000 unlabeled samples.

5 Discussion

The fine-tuning of the ParsBERT model for emotion classification on Farsi (Dari) social media texts demonstrates several significant findings and sheds light on broader implications for sentiment analysis in low-resource settings.

5.1 Model Performance and Findings

The model achieved a validation accuracy of 53.89% and a weighted F1-score of 52.17%. While these metrics might appear modest in comparison to high-resource language models, I personally observed that zero-shot classification with multilingual models was ineffective—most texts were labeled as 'neutral'—so this improvement with fine-tuned ParsBERT was a significant step forward in my project, which labeled over 95% of the texts as neutral. This clearly indicates that fine-tuning on culturally and linguistically relevant data significantly enhances model performance.

In terms of category-wise performance, common sentiments such as *happiness*, *sadness*, and *anger* were more accurately classified, while minority categories such as *fear*, *disgust*, and *hope* exhibited greater confusion and lower precision. The inclusion of *hope* as a new emotion class proved beneficial in capturing a range of positive, forward-looking comments that would otherwise be misclassified as *neutral* or *happiness*.

5.2 Significant Insights

- **Dari-Specific Challenges:** The model's performance validated the hypothesis that dialectal variations, informal tone, and mixed-script input (e.g., Latin transliterations) present considerable challenges for off-the-shelf NLP models.
- **Fine-Tuning Value:** ParsBERT, despite being pretrained on Persian corpora, benefitted greatly from fine-tuning on real-world Dari text, supporting the view that domain-specific adaptation is essential for low-resource languages.
- **Emotion Label Utility:**From my manual labeling experience, I realized that many comments didn't fit neatly into standard categories. Adding hope helped me capture encouraging and optimistic tones, especially common in diaspora conversations.

5.3 Limitations

Despite promising results, several limitations remain:

- **Label Imbalance:** The training dataset exhibited class imbalance, with emotions like *fear*, *disgust*, and *surprise* underrepresented. This skew may have affected the model's ability to generalize well.
- **Manual Labeling Scope:** Only about 4,000 comments were manually labeled, which may limit the diversity of emotional expressions encountered during training.
- **Colloquial Variance:** From my experience labeling and testing the data, I noticed that informal or sarcastic comments—especially jokes or political satire—often confused the model and led to misclassifications.

5.4 Future Work

This project sets a strong foundation for further advancements in Persian (Dari) sentiment analysis. Potential extensions include:

- **Larger Labeled Corpus:** Expanding the manually labeled dataset to 10,000+ examples would improve robustness and support more granular classification.
- **Data Augmentation:** Leveraging synonym substitution or back-translation could mitigate class imbalance and improve generalization.
- **Multi-Task Learning:** Incorporating POS tagging or Named Entity Recognition in a multi-task setup could improve the model's linguistic understanding.

- **Dialect Adaptation:** Future work could include separate training for regional variants of Dari (e.g., Herati, Hazaragi) to improve dialect sensitivity.
- **Platform-Based Customization:** Analyzing differences between Telegram and YouTube comments may enable platform-specific tuning, improving performance in practical applications.
- **Explainability:** Adding attention visualization or SHAP analysis would provide interpretability, aiding in trust and usability for social science or policy research.

Overall, the combination of domain-informed dataset construction, culturally relevant emotion classes, and transformer-based fine-tuning presents a replicable pathway for emotion analysis in other low-resource and dialectally rich languages.

6 Conclusion

This study presents a comprehensive and practice-driven approach to sentiment and emotion analysis of Farsi (Dari) social media content. By manually collecting, labeling, and fine-tuning the ParsBERT model on real-world comments from Telegram and YouTube, I was able to achieve significant improvements over zero-shot models—especially in recognizing nuanced sentiments like "hope," which are often overlooked in conventional emotion taxonomies.

The project highlights how cultural and linguistic adaptation is essential in NLP for underrepresented languages. My key contributions include the creation of a cleaned, annotated dataset reflecting Afghan discourse, the thoughtful extension of emotion labels with a culturally relevant category, and the successful implementation of a fine-tuned transformer model tailored to Dari content.

Through this process, I gained firsthand insight into the complexities of social media language, including dialectal diversity, sarcasm, and transliterated text. These results provide a solid foundation for advancing NLP research and applications in Persian and Dari. Looking ahead, this work can be extended by increasing the volume of labeled data, addressing class imbalance, and incorporating dialect- or platform-specific customization to support even more accurate and inclusive emotion modeling.

7 References

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